

Fractional Knapsack

Greedy Approach

$O(n \log n)$
Since
Sorting
by P/w ratio.

Total weight = 50

profit = [60, 100, 120]

weights = [10, 20, 30]

Step 1: Calculate profit/ratio for each item.

P $\rightarrow$ 60, 100, 120

W $\rightarrow$ $\frac{10}{6}$, $\frac{20}{5}$, $\frac{30}{4}$

Step 2: Sort elements based on P/w ratio (highest to lowest)

Step 3:

Step	Action	Capacity Left	value	Total
1	pick item 1 (60)	50 - 10 = 40	+60	60
2.	pick item 2 (100)	40 - 20 = 20	+100	160
3.	since 3rd item weight 30 > 20 take 20/30	20/30 = 0.66	0.66x 120 ≈ 80	160 + 80 <u>240</u>